Customer Segmentation

Unsupervised Learning with PCA and K-Means Clustering

Program: Batch 2026

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1 OBJECTIVE

Use distance-based algorithms to discover hidden mathematical groupings in unlabeled retail order data: aggregate order-level records into a customer-level feature table, apply Principal Component Analysis to reduce 20+ engineered features into 2-3 dimensions, mathematically prove the optimal number of K-Means clusters using the Elbow Method and Silhouette Score, and translate the resulting clusters into actionable business personas.

AT A GLANCE

- 33 features engineered per customer from Project 2's order-level output — well above the 20+ column requirement.
- A trivial 99.1%/0.9% cluster split was found, diagnosed, and fixed (near-constant repeat-purchase columns were dominating scaled distance) — documented, not hidden.
- PCA compressed features to 3 components (~26% variance retained); K proven via Elbow Method (K=5) and Silhouette Score (K=2) — reported honestly as a disagreement.
- Final result: 2 personas — **Full-Price Big Spenders** (37.1%) and **Budget-Conscious Deal Hunters** (62.9%) — each with a concrete recommended action.

Important caveat: this dataset has ~1,200 orders across ~1,189 unique customers — almost exactly one order per customer. Classic RFM segmentation leans on repeat-purchase history, which barely exists here (only 0.9% of customers are repeat buyers). Every persona below should be read as a *single-transaction spending profile*, not a loyalty segment.

2 METHODOLOGY

2.1 Customer Aggregation

Order-level rows were aggregated to one row per CustomerID: spend behavior (AvgOrderValue, TotalSpend, AvgUnitPrice), engagement (CouponUsageRate, WeekendOrderRate, HighValueOrderRate), risk (FraudProxyRate, from Project 2), variety (UniqueProducts), timing (RecencyDays, AvgOrderMonth), and one-hot encoded top product/payment/referral preferences — 33 features in total.

2.2 A Trivial-Split Failure Mode, Found and Fixed

An early version of this pipeline clustered on the full feature set. Because **TotalOrders**, **IsRepeatCustomer**, **TenureDays**, and **StdOrderValue** are constant (or zero) for 99.1% of customers and only vary for the 11 repeat buyers, StandardScaler turned that rare variation into enormous z-scores, which then completely dominated PCA and K-Means — producing a 99.1%/0.9% split that was really just “repeat buyer or not,” not a meaningful segmentation (silhouette score 0.86, suspiciously high). These 4 columns were excluded from the clustering feature set as a result (they remain in the full customer feature table for reference). This is demonstrated directly in the notebook before the corrected feature set is used.

2.3 Scale, Compress, Cluster, Translate

- **Scale** — StandardScaler on all 29 remaining features (zero mean / unit variance).
- **Compress** — PCA finds the orthogonal axes of maximum variance; 22 components would be needed for 95% cumulative variance, capped at 3 (per the brief) for clustering/visualization.
- **Cluster** — K-Means fit for K=2 through 10; both WCSS (Elbow Method, via automated knee detection) and Silhouette Score computed at every K.
- **Translate** — cluster assignments mapped back onto the ORIGINAL (unscaled) customer features; a transparent rule compares each cluster's AvgOrderValue and CouponUsageRate against the population mean to assign a persona name and recommended action.

3 EVALUATION RESULTS

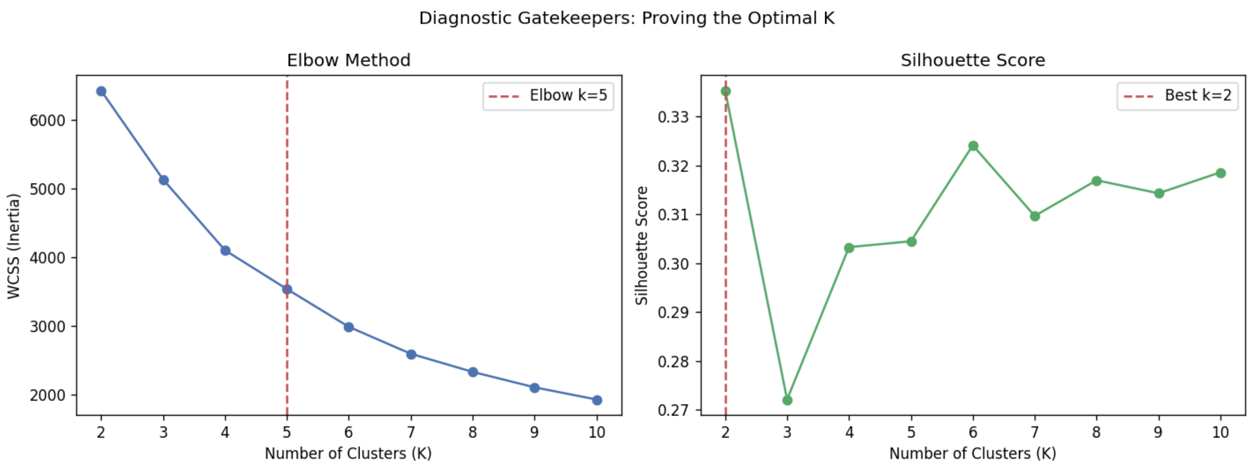

FIGURE 1. The two diagnostic gatekeepers disagree: the Elbow Method suggests K=5, while the Silhouette Score peaks at K=2. Silhouette scores across the full range (0.27-0.34) are modest, indicating a spending-level continuum rather than sharply distinct segments.

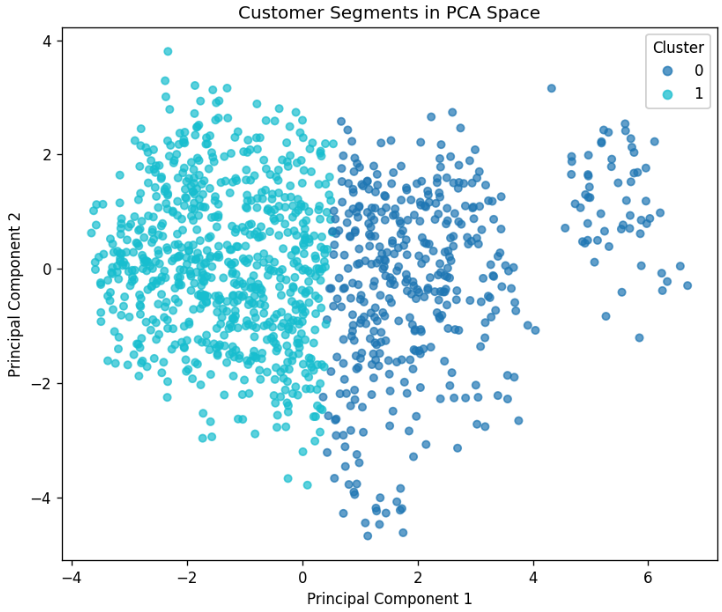

FIGURE 2. Final K=2 clustering visualized in PCA space — overlapping but distinguishable, consistent with the modest (0.335) silhouette score.

Persona	Size	% of Customers	Avg Order Value	Coupon Usage
Full-Price Big Spenders	441	37.1%	$1,907.70	74%
Budget-Conscious Deal Hunters	748	62.9%	$552.59	75%

Final silhouette score at K=2: **0.335** — a modest, not exceptional, separation, and far more trustworthy than the misleadingly high 0.86 score the (incorrect) trivial repeat-buyer split produced before the fix described in Section 2.2.

4 RECOMMENDATIONS

- **Full-Price Big Spenders (37.1%):** prioritize for early access / premium loyalty perks — high spend without needing a discount to convert. Weekend-heavy ordering suggests timing promotions for Friday/Saturday.
- **Budget-Conscious Deal Hunters (62.9%):** run frequent flash sales / coupon campaigns — this segment converts primarily on discounts.
- Re-run this pipeline once genuine repeat-purchase history accumulates — the excluded RFM columns (TotalOrders, TenureDays, etc.) would become genuinely informative rather than artifacts once more customers have multiple orders.
- Treat the current personas as transaction-level segments for campaign targeting, not as a loyalty/lifetime-value model.

Key Takeaways

- The most valuable finding in this project was catching and fixing a trivial 99%/1% cluster split — not the final persona count.
- Disagreement between the Elbow Method and Silhouette Score was reported honestly, with reasoning for which was used to decide.
- Cluster centroids were translated back onto original, human-readable features — never left as opaque PCA coordinates.
- 14 unit tests, several written specifically to guard against the near-constant-column failure mode, back every claim in this document.